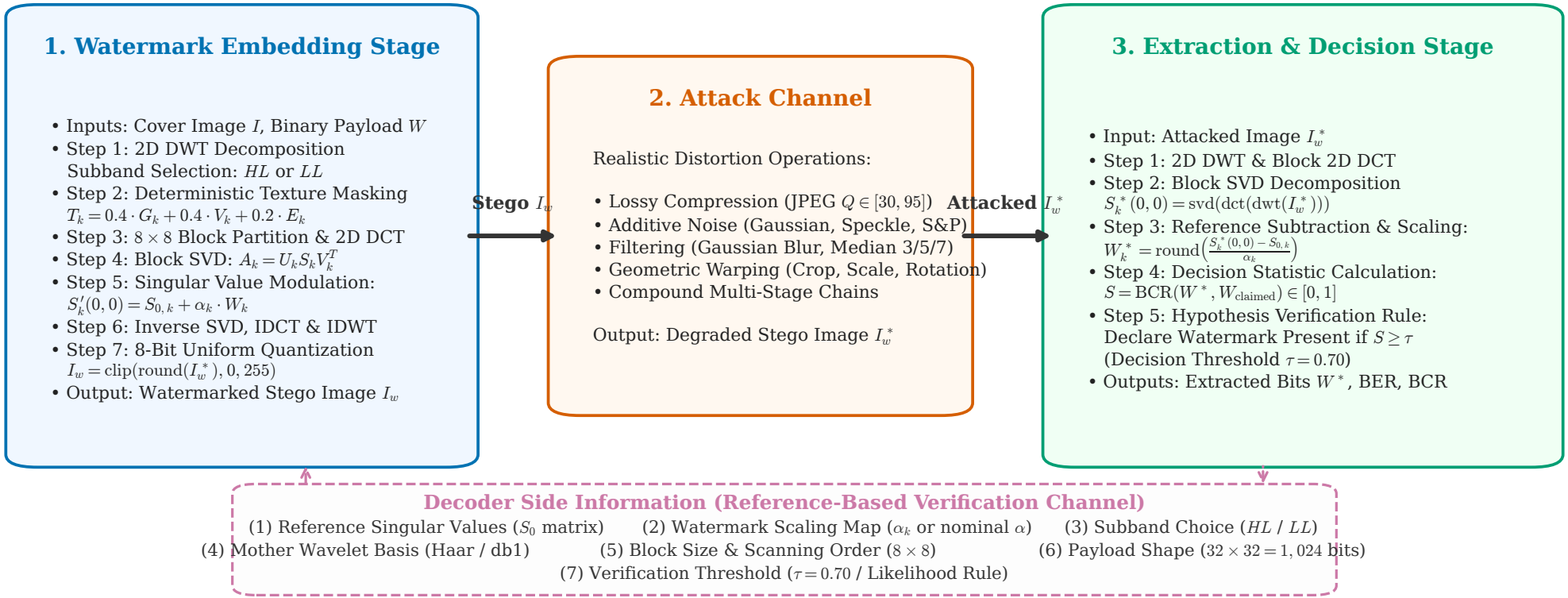

1. Watermark Embedding Stage
• Inputs: Cover Image I, Binary Payload W
• Step 1: 2D DWT Decomposition
Subband Selection: HL or LL
• Step 2: Deterministic Texture Masking
$T_k = 0.4 \cdot G_k + 0.4 \cdot V_k + 0.2 \cdot E_k$
• Step 3: 8×8 Block Partition & 2D DCT
• Step 4: Block SVD: $A_k = U_k S_k V_k^T$
• Step 5: Singular Value Modulation:
$S'_k(0,0) = S_{0,k} + \alpha_k \cdot W_k$
• Step 6: Inverse SVD, IDCT & IDWT
• Step 7: 8-Bit Uniform Quantization
$I_w = \text{clip}(\text{round}(I_w^*), 0, 255)$
• Output: Watermarked Stego Image I_w
Stego I_w
2. Attack Channel
Realistic Distortion Operations:
• Lossy Compression (JPEG $Q \in [30, 95]$)
• Additive Noise (Gaussian, Speckle, S&P)
• Filtering (Gaussian Blur, Median 3/5/7)
• Geometric Warping (Crop, Scale, Rotation)
• Compound Multi-Stage Chains
Output: Degraded Stego Image I_w^*
Attacked I_w^*
3. Extraction & Decision Stage
• Input: Attacked Image I_w^*
• Step 1: 2D DWT & Block 2D DCT
• Step 2: Block SVD Decomposition
$S_k^*(0,0) = \text{svd}(\text{dct}(\text{dwt}(I_w^*)))$
• Step 3: Reference Subtraction & Scaling:
$W_k^* = \text{round}\left(\frac{S_k^*(0,0) - S_{0,k}}{\alpha_k}\right)$
• Step 4: Decision Statistic Calculation:
$S = \text{BCR}(W^*, W_{\text{claimed}}) \in [0, 1]$
• Step 5: Hypothesis Verification Rule:
Declare Watermark Present if $S \geq \tau$
(Decision Threshold $\tau = 0.70$)
• Outputs: Extracted Bits W^*, BER, BCR
Decoder Side Information (Reference-Based Verification Channel)
(1) Reference Singular Values (S_0 matrix)
(2) Watermark Scaling Map (α_k or nominal α)
(3) Subband Choice (HL / LL)
(4) Mother Wavelet Basis (Haar / db1)
(5) Block Size & Scanning Order (8×8)
(6) Payload Shape ($32 \times 32 = 1,024$ bits)
(7) Verification Threshold ($\tau = 0.70$ / Likelihood Rule)